BLE Link Performance Demo - Results Report

MATLAB R2022b Update 3 (Bluetooth, Communications, Signal Processing, Antenna toolboxes; no WLAN or RF Toolbox). All values are **simulated** and were printed by the MATLAB runs (results/*.mat, exported to results/report_data.json). Fixed random seeds.

Summary

- Sensitivity at 30.8% PER (ideal receiver, 7 dB NF, 1000 packets/point): -100.2 dBm (LE1M), -97.2 (LE2M), -104.1 (LE500K), -108.3 (LE125K).
- Noncoherent (discriminator) receiver: needs 8.81 dB Eb/No at 30.8% PER and keeps PER at 5.0% at the 150 kHz CFO spec limit; the toolbox ideal receiver fails from 2.5 kHz.
- Indoor multipath (Step 8): fading penalty at 30.8% PER vs AWGN is 3.8 / 4.3 dB (LE1M, Rician / Rayleigh) and 2.8 / 3.9 dB (LE125K).
- Impairment cost (Step 9, LE1M): 150 kHz CFO 1.27 dB, -70 dBc/Hz phase noise 1.85 dB, 30% DC offset 0.98 dB, 50 ppm timing drift 0.01 dB.
- Receiver test suite at RF-PHY spec levels (Step 10): 2 regressions caught; intermodulation fails once IIP3 drops below -37.5 dBm.
- Ray-traced coverage in the shipped 'office' scene (Step 12): LE1M covers 99.1%, LE2M 90.8%, LE125K 100.0% of points reached by rays at -20 dBm.
- Bursty Wi-Fi with a learning AFH (Step 13): PER 18.50% without AFH vs 7.00% with it over 2000 events; all overlapped channels excluded by event 737, map final at event 1751.

Code added in this round

- `blePerSim.m` options (defaults unchanged, verified bit-identical on 4 regression cases): `Channel` (awgn / rician / rayleigh), `TimingDrift_ppm`, `PhaseNoise_dBcHz`, `DCOffset`, `Intf_Type` (ble / cw / twotone / wifi) with `Intf_Offset_Hz`, `Intf_CI_dB`, `Intf_BW_Hz`, `Intf_WifiMask`, and absolute levels `WantedLevel_dBm` + `IIP3_dBm` (comm.MemorylessNonlinearity).
- New scripts `step8_multipath.m` ... `step13_coexistence.m`; helper `wifiOverlap.m` shared by Steps 1 and 6; Step 2 now 1000 packets/point; `run_all.m` runs all 13 steps.
- `report/export_results_json.m` + `report/make_report.py` regenerate this PDF from the results.

Resume claims and supporting evidence

The resume bullet wording was not provided, so the claims below are the capabilities this work was asked to demonstrate. Replace them with the actual bullets and check each against the evidence column.

Claim	Supported	Evidence (figure / number)
Link-level BLE PHY simulator for all 4 LE PHYs (PER, sensitivity)	Yes	Fig. Step 2; sensitivity -100.2 / -108.3 dBm (LE1M / LE125K), 1000 pkts/point
Realistic noncoherent receiver tolerant to +/-150 kHz CFO	Yes	Fig. Step 3; PER 5.0% at 150 kHz; 8.81 dB vs 6.77 dB ideal
Indoor multipath fading (Rician/Rayleigh) and its link penalty	Yes	Fig. Step 8; 4.3 dB (LE1M, Rayleigh)
Quantified RF impairment budget (CFO, drift, phase noise, DC)	Yes (phase noise partial)	Fig./table Step 9; comm.PhaseNoise accuracy warning at -75/-70 dBc/Hz
Automated receiver conformance suite (C/I, blocking, intermod) with regression detection	Partial	Fig. Step 10; 2 regressions; blocking passes trivially (ideal filter, no reciprocal mixing)
Front-end linearity requirement from intermod test	Yes	Step 10b: fails below IIP3 -37.5 dBm (nominal assumed -20 dBm)
Transmitter tests (power, modulation, CFO/drift) vs spec limits	Partial	Table Step 11; toolbox waveform has no ISI so df2/df1 ratio is not representative; no PA model
Ray-traced indoor coverage and PHY selection	Yes	Fig. Step 12; office scene, 584 of 640 grid points reached; LE2M out-delivers LE125K at 100% of them
Wi-Fi coexistence with adaptive frequency hopping	Yes	Fig. Step 13; PER 18.50% -> 7.00%; 13 channels excluded

Baseline demo (Steps 1-7)

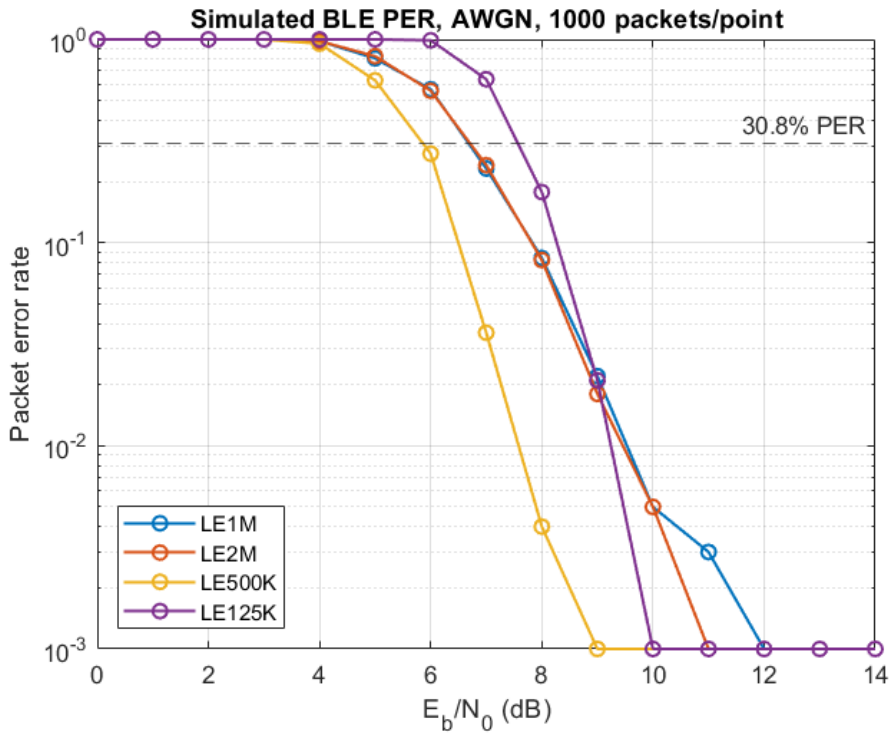
Waveforms and hopping (Step 1): Wi-Fi channel 6 blocks 10 of 37 data channels (shared `wifiOverlap` rule); CSA #1 landed in the Wi-Fi band 27/100 times without AFH and 0/100 with AFH.

PHY	Bit rate (kbps)	Eb/No at 30.8% PER (dB)	Sensitivity (dBm)
LE1M	1000	6.77	-100.2
LE2M	2000	6.79	-97.2
LE500K	500	5.90	-104.1
LE125K	125	7.72	-108.3

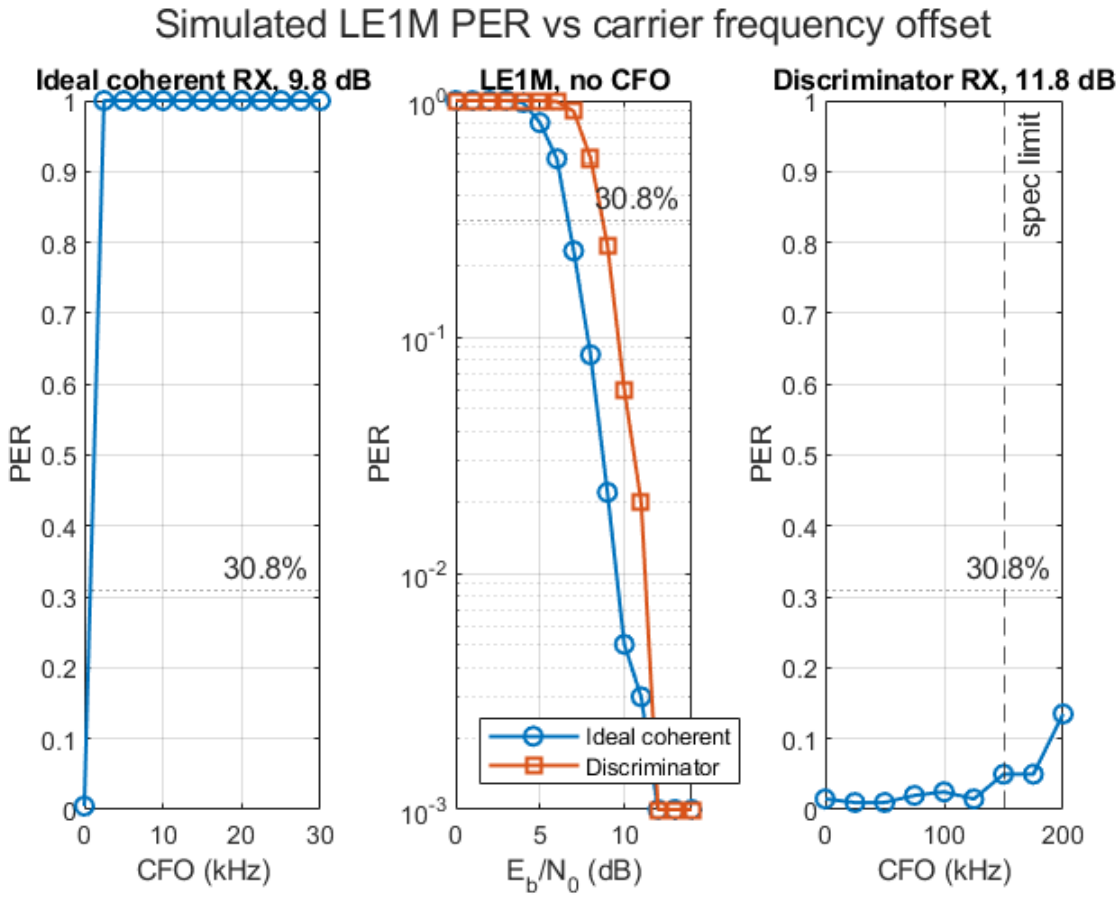
Step 2 (ideal receiver, 1000 packets/point). Step 3: discriminator receiver needs 8.81 dB; CFO sweep at 11.8 dB gives 0 kHz: 1.5%, 25 kHz: 1.0%, 50 kHz: 1.0%, 75 kHz: 2.0%, 100 kHz: 2.5%, 125 kHz: 1.5%, 150 kHz: 5.0%, 175 kHz: 5.0%, 200 kHz: 13.5%.

Step 4 (toy log-distance floor plan): range LE1M 21.9 m, LE2M 17.4 m, LE500K 29.5 m, LE125K 40.8 m; coverage LE1M 51.6%, LE2M 41.1%, LE500K 66.3%, LE125K 66.9%.

Step 5: 2 regressions (sensitivity tests at 10.8 / 12.7 dB). Step 6 (analytic Wi-Fi, 50% duty): PER 13.5% without AFH, 0.0% with AFH, 10 overlapped channels.



Step 2. PER waterfall, AWGN, ideal receiver.



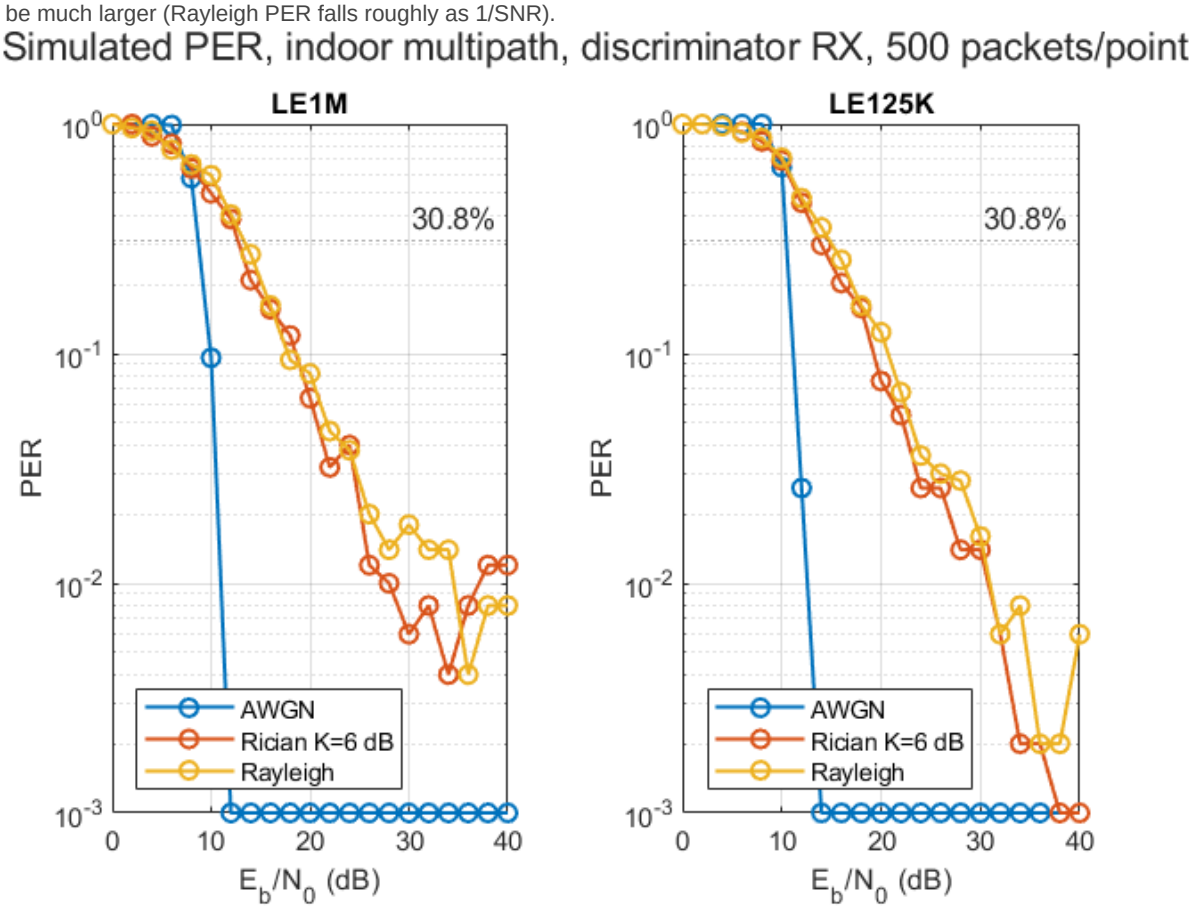
Step 3. CFO: ideal vs discriminator receiver.

Step 8 - Multipath fading

Indoor profile: delays 0/50/100/150 ns, gains 0/-3/-6/-9 dB, Rician K = 6 dB, 3 Hz Doppler, new realization per packet (block fading). Discriminator receiver, 500 packets/point, Eb/No grid step 2 dB. The largest path delay (150 ns) is much shorter than the 1 us symbol, so this is flat fading: the loss comes from whole-packet fades, not ISI.

PHY	Channel	Eb/No at 30.8% PER (dB)	Penalty vs AWGN (dB)
LE1M	awgn	9.13	0.00
LE1M	rician	12.88	3.76
LE1M	rayleigh	13.44	4.31
LE125K	awgn	11.10	0.00
LE125K	rician	13.87	2.77
LE125K	rayleigh	14.95	3.85

Note: the AWGN LE1M threshold here (9.13 dB) is higher than Step 3's 8.81 dB because linear interpolation on the coarser 2 dB grid overestimates it; penalties use the same grid on both sides. At 30.8% PER the fading penalty is modest; at lower PER targets it would be much larger (Rayleigh PER falls roughly as 1/SNR).



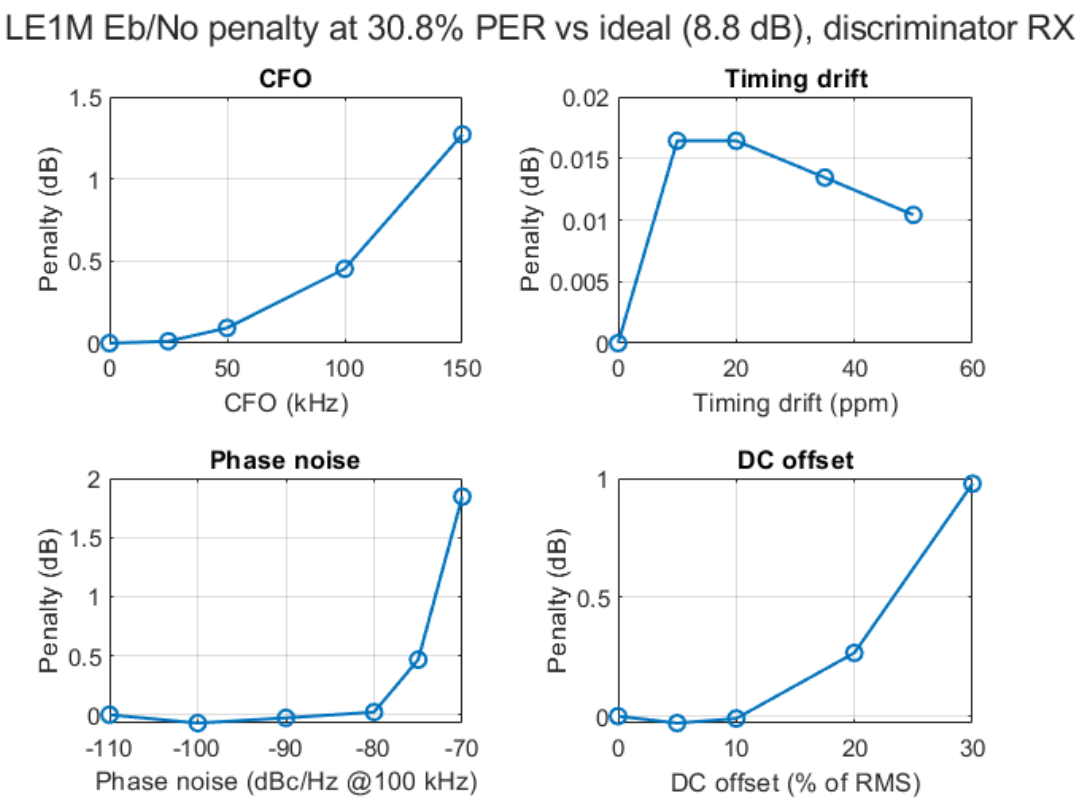
Step 8. PER vs Eb/No in AWGN, Rician and Rayleigh channels.

Step 9 - RF impairment cost table

LE1M, AWGN, discriminator receiver, 500 packets/point. Unimpaired threshold 8.85 dB Eb/No at 30.8% PER; penalty = threshold with the impairment minus that.

Impairment	Value	Unit	Eb/No req (dB)	Penalty (dB)
CFO	0	kHz	8.85	0.00
CFO	25	kHz	8.86	0.01
CFO	50	kHz	8.94	0.09
CFO	100	kHz	9.30	0.45
CFO	150	kHz	10.12	1.27
Timing drift	0	ppm	8.85	0.00
Timing drift	10	ppm	8.86	0.02
Timing drift	20	ppm	8.86	0.02
Timing drift	35	ppm	8.86	0.01
Timing drift	50	ppm	8.86	0.01
Phase noise	-100	dBc/Hz @100 kHz	8.77	-0.07
Phase noise	-90	dBc/Hz @100 kHz	8.82	-0.03
Phase noise	-80	dBc/Hz @100 kHz	8.87	0.02
Phase noise	-75	dBc/Hz @100 kHz	9.31	0.47
Phase noise	-70	dBc/Hz @100 kHz	10.69	1.85
DC offset	0	% of RMS	8.85	0.00
DC offset	5	% of RMS	8.82	-0.03
DC offset	10	% of RMS	8.84	-0.01
DC offset	20	% of RMS	9.11	0.27
DC offset	30	% of RMS	9.82	0.98

Penalties within about +/-0.1 dB of zero (including small negative values) are Monte-Carlo noise at this packet count. comm.PhaseNoise warned that its -75 and -70 dBc/Hz settings may be inaccurate. Timing drift up to 50 ppm is harmless: over a 336 us packet it slips the sampling instant by a small fraction of a sample.



Step 9. Eb/No penalty vs impairment level.

Step 10 - Receiver interference tests

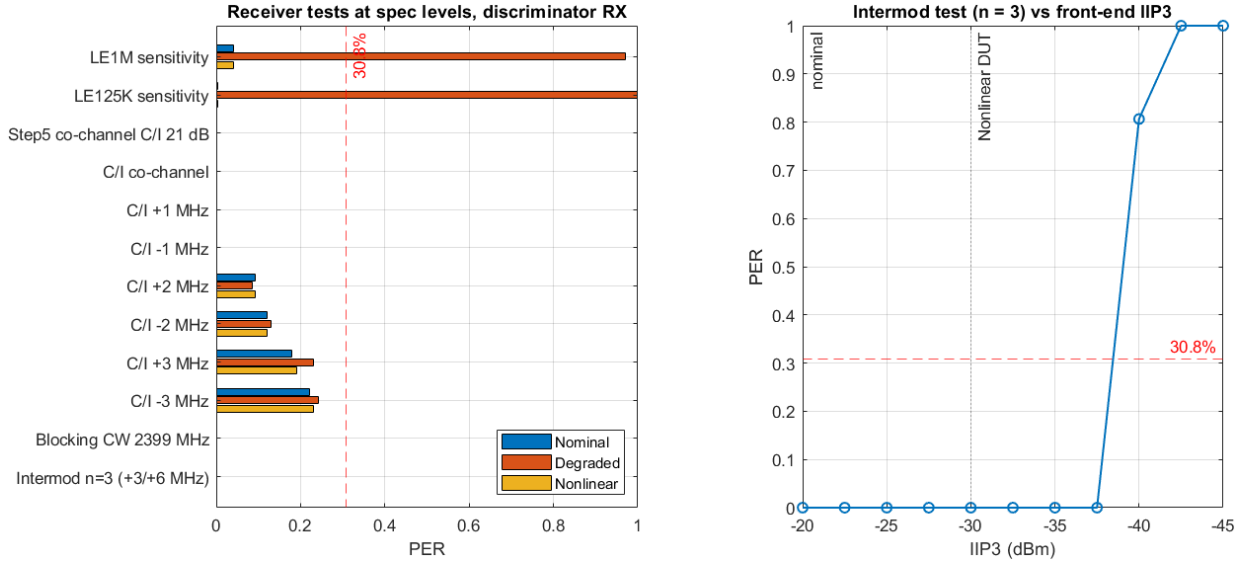
Levels from the MathWorks example 'Bluetooth LE Blocking, Intermodulation and Carrier-to-Interference Performance Tests' (`helperBLETestParamGenerate.m`, RF-PHY TS 4.5), cross-checked against `bluetoothRFPHYTestConfig`: wanted -67 dBm; BLE interferer -88 (co-channel), -82 (+/-1 MHz), -50 (+/-2 MHz), -40 dBm (+/-3 MHz); CW blocker -35 dBm at 2399 MHz (Fc 2426); intermod wanted -64 dBm with CW at +3 MHz and BLE at +6 MHz, both -50 dBm. PER limit 30.8%. DUTs: Nominal (IIP3 -20 dBm, assumed), Degraded (150 kHz CFO + 4 dB loss), Nonlinear (IIP3 -30 dBm). 500 packets per test.

Test	Nominal	Degraded	Nonlinear
LE1M sensitivity	PASS (4.0%)	FAIL (97.1%)	PASS (4.0%)
LE125K sensitivity	PASS (0.4%)	FAIL (100.0%)	PASS (0.4%)
Step5 co-channel C/I 21 dB	PASS (0.0%)	PASS (0.0%)	PASS (0.0%)
C/I co-channel	PASS (0.0%)	PASS (0.0%)	PASS (0.0%)
C/I +1 MHz	PASS (0.0%)	PASS (0.0%)	PASS (0.0%)
C/I -1 MHz	PASS (0.0%)	PASS (0.0%)	PASS (0.0%)
C/I +2 MHz	PASS (9.2%)	PASS (8.6%)	PASS (9.2%)
C/I -2 MHz	PASS (12.0%)	PASS (13.0%)	PASS (12.0%)
C/I +3 MHz	PASS (17.8%)	PASS (23.0%)	PASS (19.0%)
C/I -3 MHz	PASS (22.1%)	PASS (24.2%)	PASS (23.1%)
Blocking CW 2399 MHz	PASS (0.0%)	PASS (0.0%)	PASS (0.0%)
Intermod n=3 (+3/+6 MHz)	PASS (0.0%)	PASS (0.0%)	PASS (0.0%)

Regressions (PASS on Nominal, FAIL elsewhere): Degraded: LE1M sensitivity; Degraded: LE125K sensitivity. Both are the Degraded DUT's sensitivity tests: its 4 dB loss exceeds the 2 dB test margin (Step 5 isolated this). The Nonlinear DUT passes everything: with -50 dBm tones and IIP3 -30 dBm, IM3 lands near -90 dBm, well below the -64 dBm wanted signal.

IIP3 (dBm)	-20.0	-22.5	-25.0	-27.5	-30.0	-32.5	-35.0	-37.5	-40.0	-42.5	-45.0
Intermod PER	0%	0%	0%	0%	0%	0%	0%	0%	81%	100%	100%

Blocking passes on every DUT for a modelling reason: the CW blocker is 27 MHz away and is removed by the ideal digital channel filter, thermal noise is input-referred (compression scales noise and signal together), and LO reciprocal mixing is not modelled. Adjacent C/I at +/-3 MHz has the least margin; its PER comes from the finite selectivity of the receiver's FIR channel filter.



Step 10. Test PER per DUT, and intermod PER vs IIP3.

Step 11 - Transmitter tests

`bluetoothRFPHYTestConfig` + `bluetoothTestWaveform` with the measurement helpers from the MathWorks transmitter-test example (helperBLEModulationTestConfig / helperBLEModulationTestMeasurements); limits from its helperBLEModulationTestVerdict. Impaired TX: 60 kHz CFO + 30 kHz linear drift over the packet.

TX	Measurement	Value	Limit	Verdict
Nominal	Mod: df1avg (kHz)	250	225..275	PASS
Nominal	Mod: df2max > 185 kHz (% of samples)	100	>= 99.9	PASS
Nominal	Mod: df2avg/df1avg	1.18	> 0.8	PASS
Nominal	CFO: |f0| (kHz)	0	<= 150	PASS
Nominal	Drift: max |fn - f0| (kHz)	0	<= 50	PASS
Nominal	Drift: |f1 - f0| (kHz)	0	<= 23	PASS
Nominal	Drift: max |fn - fn-5| (kHz)	0	<= 20	PASS
Nominal	Power: average (dBm)	0	-20..+20	PASS
Nominal	Power: peak - average (dB)	0	<= 3	PASS
Impaired	Mod: df1avg (kHz)	250	225..275	PASS
Impaired	Mod: df2max > 185 kHz (% of samples)	100	>= 99.9	PASS
Impaired	Mod: df2avg/df1avg	1.18	> 0.8	PASS
Impaired	CFO: |f0| (kHz)	59.4	<= 150	PASS
Impaired	Drift: max |fn - f0| (kHz)	26.8	<= 50	PASS
Impaired	Drift: |f1 - f0| (kHz)	4.71	<= 23	PASS
Impaired	Drift: max |fn - fn-5| (kHz)	3.94	<= 20	PASS
Impaired	Power: average (dBm)	0	-20..+20	PASS
Impaired	Power: peak - average (dB)	0	<= 3	PASS

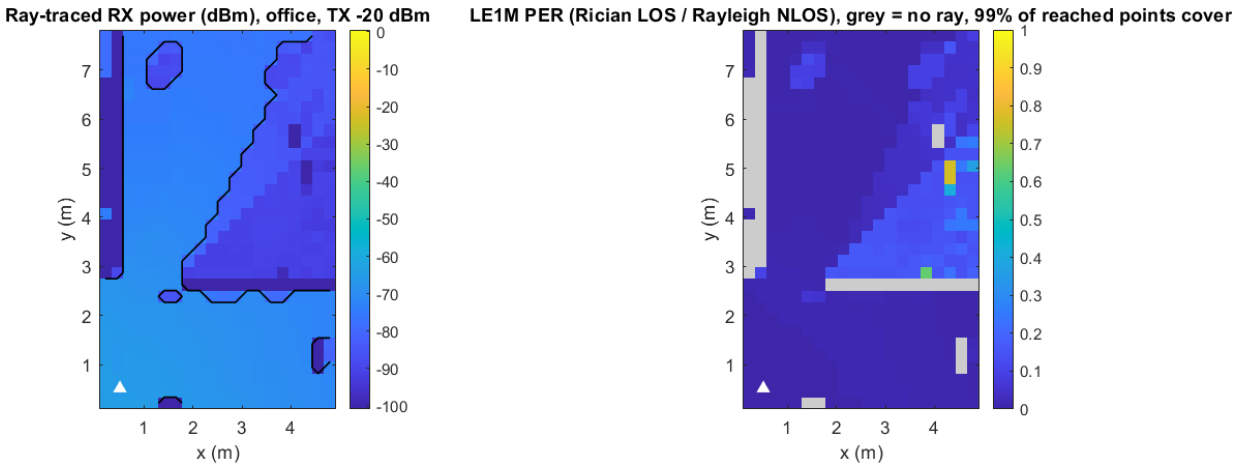
Both transmitters pass. Two caveats: (1) the toolbox GFSK uses a one-symbol frequency pulse (verified: every symbol peaks at 295 kHz for any bit pattern), so there is no ISI and the df2avg/df1avg ratio is above 1, whereas a real BT = 0.5 transmitter gives a ratio below 1; the same waveform drives all receiver results, which are therefore slightly optimistic. (2) With no PA model the constant-envelope output makes the power tests trivial. The impaired TX's drift stays inside the limits because a 30 kHz drift over a ~400 us packet is within the 50 kHz budget.

Step 12 - Ray-traced indoor coverage

Shooting-and-bouncing-ray tracing (`propagationModel("raytracing")`, up to 2 reflections, default materials) in MATLAB's shipped 'office' STL scene at 2.44 GHz. TX -20 dBm at (0.51, 0.52, 2.90) m; RX grid 0.24 m at 0.8 m height. 584 of 640 points receive rays (392 with LOS); RX power -100.8 to -64.3 dBm. PER uses Step 8's Rician curve at LOS points and Rayleigh at NLOS points; LE2M uses the LE1M curves (same GFSK, same AWGN threshold in Step 2).

PHY	Coverage, all points	Coverage, points reached by rays	Median link margin (dB)
LE1M	90.5%	99.1%	19.9
LE2M	82.8%	90.8%	16.9
LE125K	91.2%	100.0%	27.9

Points with no ray lie inside walls and furniture (straight strips and boxes in the map) and are excluded from the second coverage column. LE2M delivers more throughput than LE125K at 100% of reached points. The scene is a single 5 x 8 m office, so it does not stress the coded PHYs; a larger floor plan (e.g. a Wireless InSite export) would.



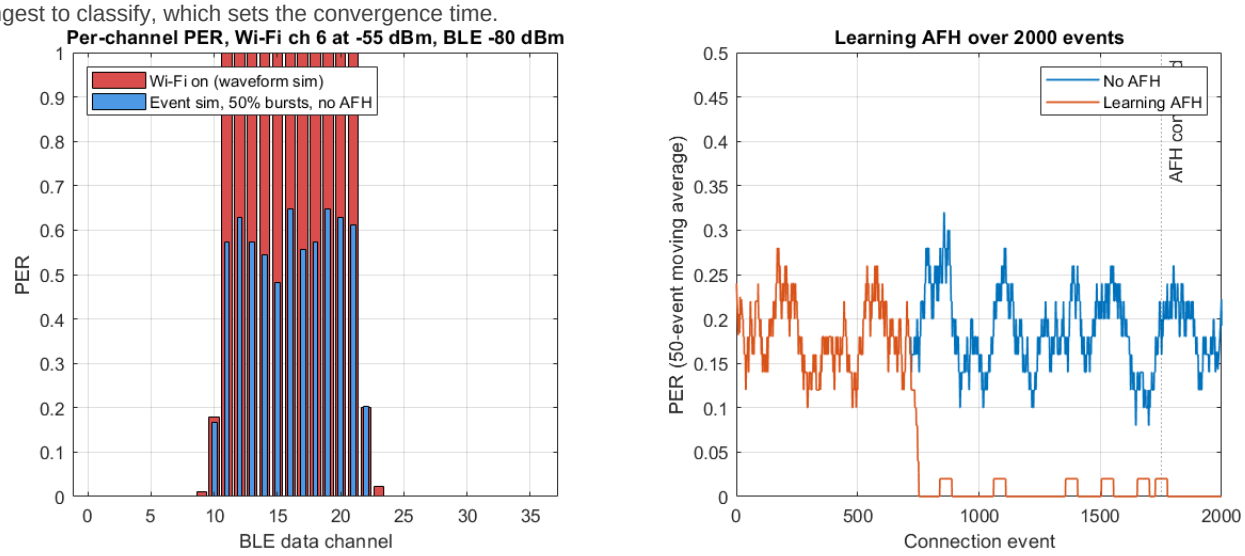
Step 12. Ray-traced RX power (black contour: LOS region) and LE1M PER map.

Step 13 - Bursty Wi-Fi and a learning AFH

Wi-Fi channel 6 at -55 dBm as Gaussian noise shaped by the 802.11 OFDM 20 MHz transmit mask (worst case allowed), wanted BLE -80 dBm, generated at 16 MHz and passed through the discriminator's channel filter with the BLE signal (500 packets per channel). Event simulation: 2000 connection events at 7.5 ms, Wi-Fi bursts 1-3 ms on/off (realised duty 49.9%), CSA #1; learning AFH marks a channel bad when its PER over its last 20 uses exceeds 20%.

Result	Value
PER, no AFH (2000 events)	18.50%
PER, learning AFH (2000 events)	7.00%
All overlapped channels excluded by event	737
Last channel-map change (converged) at event	1751
PER after convergence	0.00%
Channels excluded	10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22
Channels overlapping Wi-Fi (wifiOverlap rule)	11, 12, 13, 14, 15, 16, 17, 18, 19, 20

Partially interfered channels (outside the 20 MHz band but hit by the spectral-mask skirts): ch 9 1.2%, ch 10 18.0%, ch 21 100.0%, ch 22 20.0%, ch 23 2.2%. The AFH correctly learns to drop the worst of these as well; channels near the 20% threshold take longest to classify, which sets the convergence time.



Step 13. Per-channel PER and moving-average PER with and without learning AFH.

Not done / limitations

- Resume bullet wording not provided; the claims table uses the capabilities the work targeted.
- Blocking is not a meaningful test in this model (ideal digital filtering, input-referred noise, no LO phase-noise reciprocal mixing); the nominal IIP3 of -20 dBm is an assumption, not a datasheet value.
- The toolbox GFSK waveform has no ISI (one-symbol frequency pulse): receiver results are slightly optimistic and the TX modulation ratio is not representative of real hardware.
- comm.PhaseNoise flagged its -75 and -70 dBc/Hz settings as possibly inaccurate.
- Intermod wanted level -64 dBm follows the example helper; bluetoothRFPHYTestConfig's default shows -67 dBm.
- 500 packets per point: PER values carry roughly +/-2% statistical uncertainty; Step 8 used a 2 dB Eb/No grid.
- Ray tracing: default surface materials, 2 reflections, incoherent power sum, one small office scene; LE500K not included; LE2M borrows the LE1M fading curve.

- Coexistence: Wi-Fi is mask-shaped Gaussian noise with random bursts, not a traffic model; a packet that overlaps a burst at all uses the Wi-Fi-on PER; the AFH never re-admits channels.
- TX tests: no power-amplifier model, so output-power tests are trivially met.
- Discriminator timing is ideal (taken from the clean reference); coded-PHY path uses internal toolbox functions (ble.internal.*).